KEM cycle tester: a placeholder KEM, exercised honestly

class DummyKEM (main.py)

Holds one X25519 key pair.

encapsulate(): fresh ephemeral X25519 key, DH against the public key; the ephemeral public key is the ciphertext.

decapsulate(ciphertext): DH with the private key.

Classical X25519. Not post-quantum.

test_kem_performance(iterations, kem_type)

loops encapsulate and decapsulate, compares secrets

derive_key(secret): HKDF-SHA256, 32 bytes

encrypt and decrypt: AES-256-CBC, PKCS7

test_encryption_decryption and generate_key_pair: PEM files out

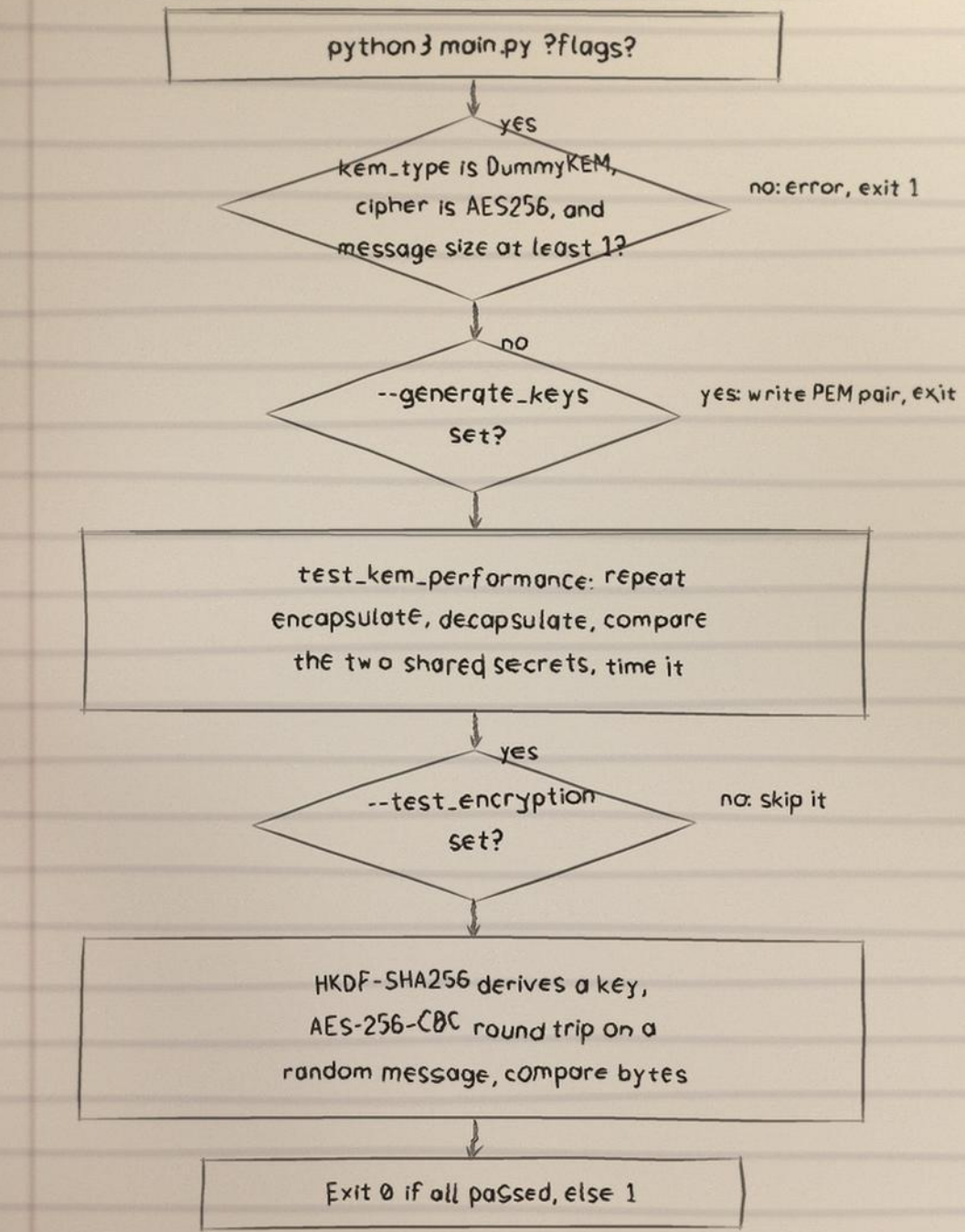
main(): validate kem_type, cipher, message size up front then keys, cycles, or the encryption test, in that order

Exit 0 only when every requested test passes. Any other kem_type or cipher is refused.

Generated key files are outputs only. No mode reads a key file back in.

folder tab, signed GK

A run of the cycle tester, gated up front



Default run: 10 iterations, no encryption test.

Product map: an interface harness, not a KEM

WHO AND WHY

Me, checking that a KEM-shaped interface round-trips correctly:
encapsulate and decapsulate must land on the same shared secret, every cycle.

WHAT IT DOES

Timed encapsulate and decapsulate cycles (default 10).
An optional AES-256-CBC round trip on a key from HKDF-SHA256.
X25519 key pair generation to PEM files.

INPUTS AND OUTPUTS

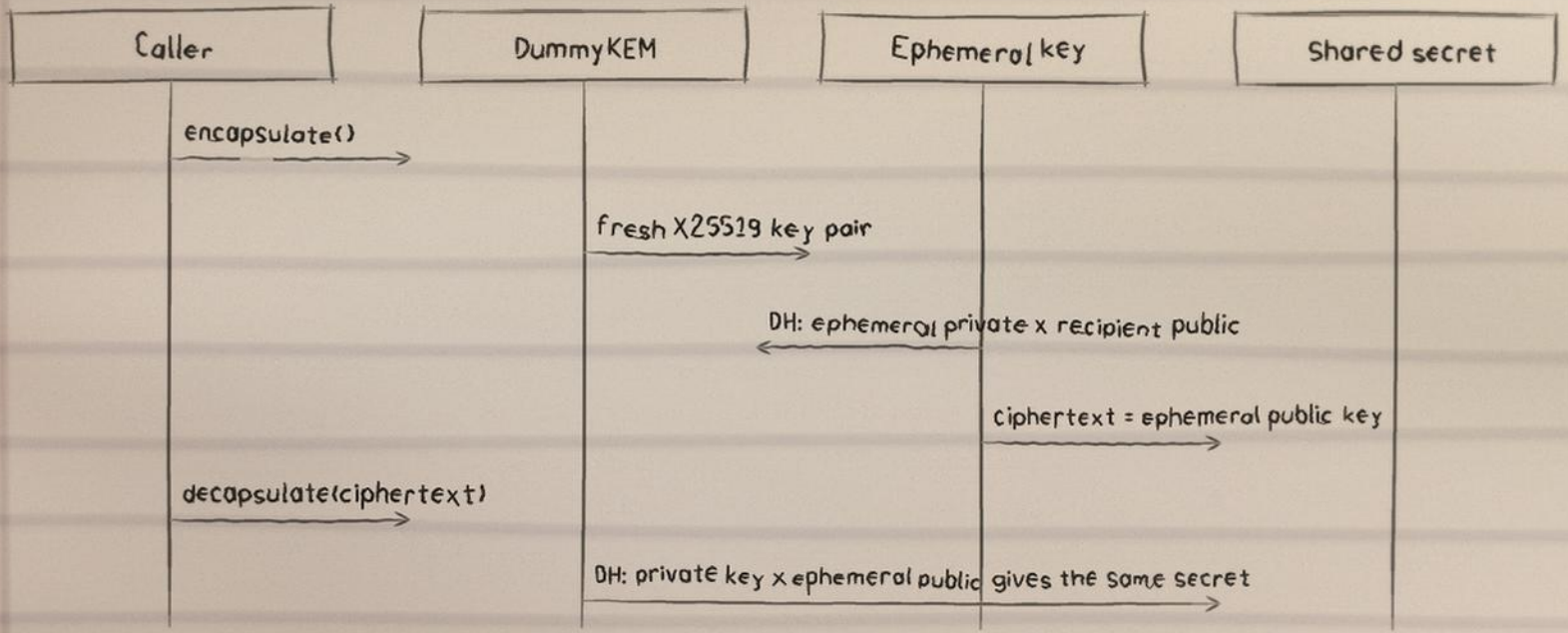
In: command line flags only. Out: log lines with timing, PEM key files when asked,
and an exit code that tells the truth about the run.

BOUNDARY, IN PLAIN WORDS

DummyKEM is X25519, a classical exchange. No post-quantum KEM is
implemented here, and this tool tests the interface, not PQ security.

signed on the band, GK

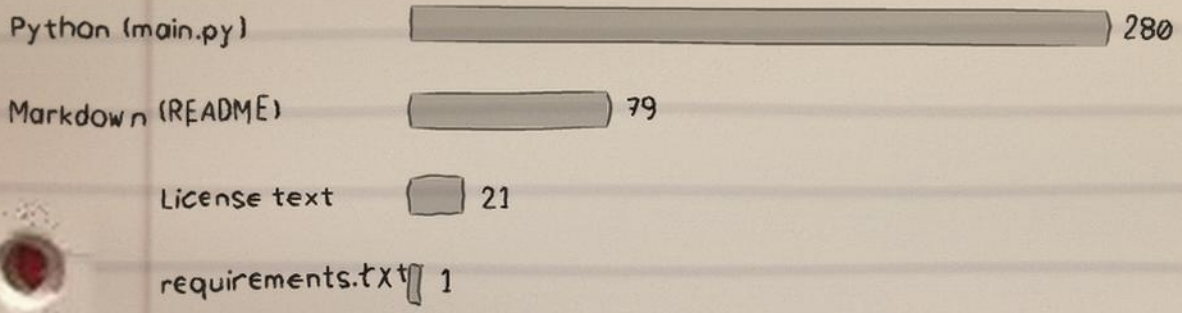
Sequence: one encapsulate and decapsulate cycle



`test_kem_performance` compares the two secrets each cycle and fails the run if they differ.

KEM cycle tester in numbers, counted 2026-10-30

Lines by kind, from git ls-files and a line count over each tracked text file.



The whole repo is 6 files and 9 commits. First commit 2025-07-22, last one 2026-10-30, both from git log.
Tests: no test suite ships in this repo, so no pass is claimed. See VISUAL.md for the run limits.
Full table with the counting method for every number: VISUAL.md in this folder.

tally kept by GK

Kem Interface Cycle Tester: schematic of the apparatus

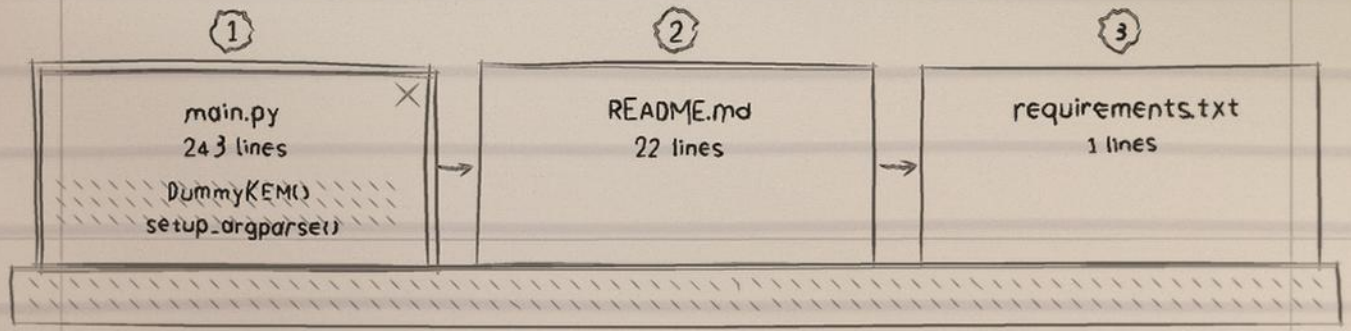
the system as a machine: parts, flow, and measurements

construction lines left in, as drawn

parts list

1. main.py
2. README.md
3. requirements.txt

in flow order, first to last



base plate: the repo itself, sectioned

span: 3 files, 266 lines counted

turns of the crank so far: 4 commits, 2025-07-22 to 2025-07-22

measured and sketched, GK